Cassava 1 — Guide

Oct 26, 2026 · Anas Alsirafy

The cassava track built a classifier for the five-class Kaggle cassava leaf-disease set under the CERERE project at AUC, and timed it on edge devices. It is also where the fine-tuning pipeline later used across all datasets was developed. Best result: EfficientNet-B0 plus a ConvNeXt-Base specialist, combined by per-class weights, at 90.32% (5-fold). This set is for a new team member with basic AI knowledge.

Where this sits

The handover has three parts. This set covers cassava. Maize has its own eight docs starting at *Cereal 1 — Guide*. The five-dataset comparison behind Paper 2, which reuses this pipeline on cassava and four other datasets, is one doc: *Cross-Dataset Study — Five Datasets (Paper 2)*.

Status

Piece	State (Oct 2026)
Classifier	Done. Best single model 87.88% (Exp 10), best ensemble 90.32% (Exp 11), 5-fold.
Edge timing	Done. Ensemble inference: RTX 5090 89 ms, Jetson Xavier NX 65 ms, Raspberry Pi 4 254 ms per image.
Field pipeline (FIWARE)	Architecture designed; how much was built beyond Jetson inference is to confirm.
Clean re-evaluation	Pending: the Exp 11 specialist was trained on folds that don't match the evaluation folds.

How to read this set

1. *Cassava 2 — Environment Setup* — hardware, packages, Kaggle and edge-device notes.
2. *Cassava 3 — Data* — the dataset, its five classes and their imbalance.
3. *Cassava 4 — Configs* — settings for the best models.
4. *Cassava 5 — Experiments* — the 11-experiment log, from 67% to 90%.
5. *Cassava 6 — Results and Findings* — what the numbers mean, including how cassava did in the cross-dataset study.
6. *Cassava 7 — Known Issues and Gotchas* — bugs and traps. Read before touching code.
7. *Cassava 8 — Future Work* — what to do next.

Glossary

Term	Meaning here
Backbone	A pretrained image network (MobileNetV2, EfficientNet-B0, ConvNeXt-Base) that turns a photo into features.
Frozen vs fine-tuned	Frozen: backbone weights locked, only a small head is trained. Fine-tuned: some or all backbone weights are trained too.

Term	Meaning here
Three-phase training	Frozen backbone → top 30 layers unfrozen → whole network unfrozen at a lower learning rate.
Noisy Student	A set of EfficientNet weights trained with self-training on extra unlabelled images; usually transfers better.
CLAHE	Local contrast boost on the lightness channel; makes lesions easier to see.
MixUp / CutMix	Blend two training images (MixUp) or paste a patch of one onto another (CutMix), mixing labels too.
Focal loss	Down-weights easy examples so training focuses on hard, usually minority-class, ones. Per-class focal loss uses a different strength (gamma) per class.
Specialist	A model trained only on the four minority classes, combined with the general model.
Weighted ensemble	Average two models' class probabilities, weighting each class by how well each model scores on it (per-class F1).
Fold / k-fold CV	Split data into 5 parts; train on 4, test on 1; repeat so every image is tested once.
Leakage	Information about test images reaching training, which inflates accuracy.
Balanced accuracy / macro-F1	Scores that weight each class equally, so ignoring the four small classes is penalised.

People and files

Who / what	Detail
Anas Alsirafy	Built this work; leaving AUC (anas.samy@aucegypt.edu)
Prof. Maki K. Habib	Supervisor; sets priorities after handover
Successor	Not yet chosen
Code and notebooks	Google Drive folder, link to be added
Dataset	Kaggle cassava leaf-disease competition (see Data)

Open questions for Anas

- [ ] Add the Google Drive link here and in Environment Setup.
- [ ] Confirm the training GPU (RTX 5090) and the Kaggle GPU model (written as P100).
- [ ] Confirm the TensorFlow version (cassava code ran on 2.19).
- [ ] Point to the records for Experiments 6–9.
- [ ] Confirm where the trained `.keras` model files are stored.
- [ ] Confirm which FIWARE links beyond Jetson inference were built.